Stint degradation

Tyre degradation from 24,386 F1 lap times, 2023, with the fuel and track-evolution effect estimated rather than assumed.

Degradation is +0.0316 s a lap [+0.0256, +0.0376]

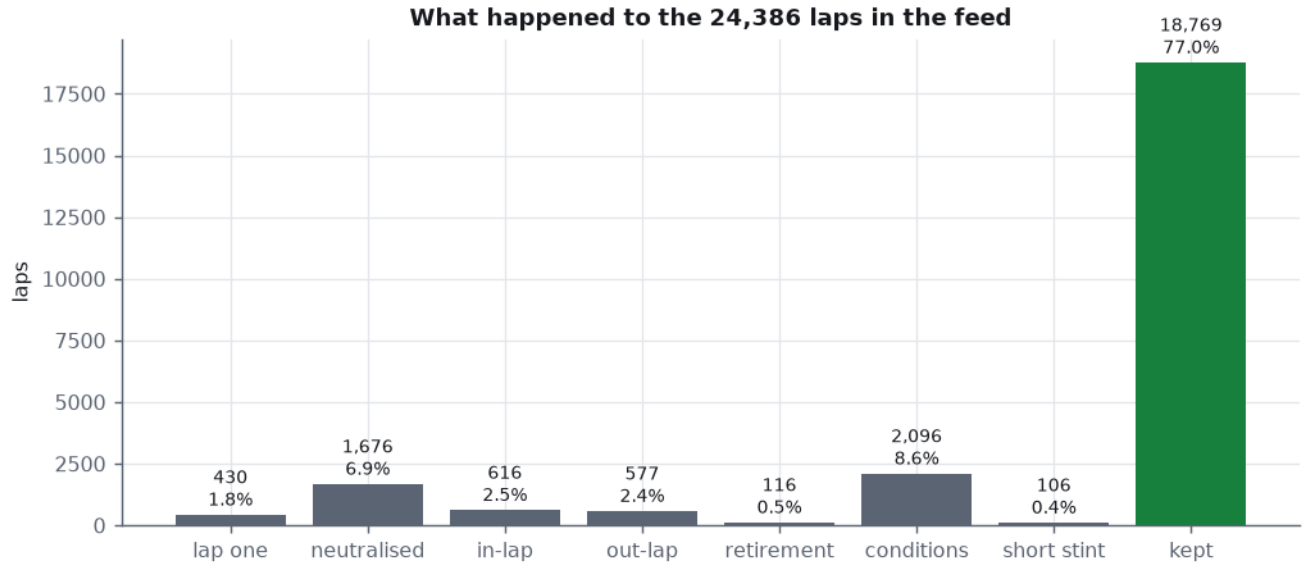
The naive estimate, leaving the race lap out, is -0.0251 — the opposite sign. From 18,769 kept laps over 384 car-races.

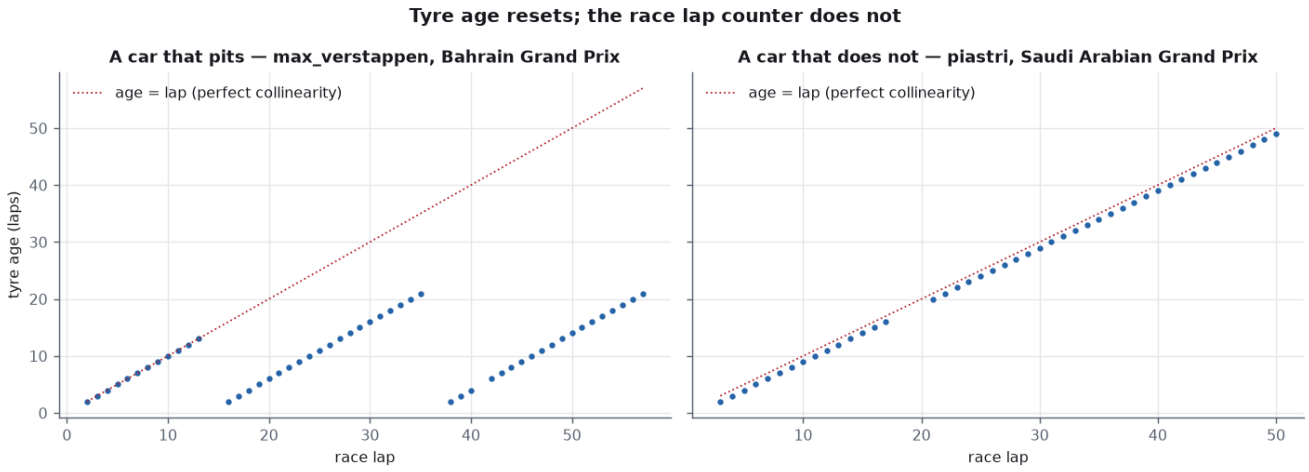
18,769	384	+0.0316	-0.0251
laps kept	car-races	degradation, s/lap	naive estimate

The whole problem in one line: within a stint, tyre age and race lap move together one for one, so leaving the race lap out does not drop the fuel effect — it hides it inside the degradation coefficient. The naive estimate is therefore not a noisier version of the right answer; it is the right answer plus the fuel effect, which here is large enough to reverse the sign.

Which laps are used

94.5% of the laps that survive belong to a car that made at least one stop — the only cars that can separate tyre age from race lap. Stints per car-race: 1 stint: 33, 2 stints: 140, 3 stints: 183, 4 stints: 27, 6 stints: 1.



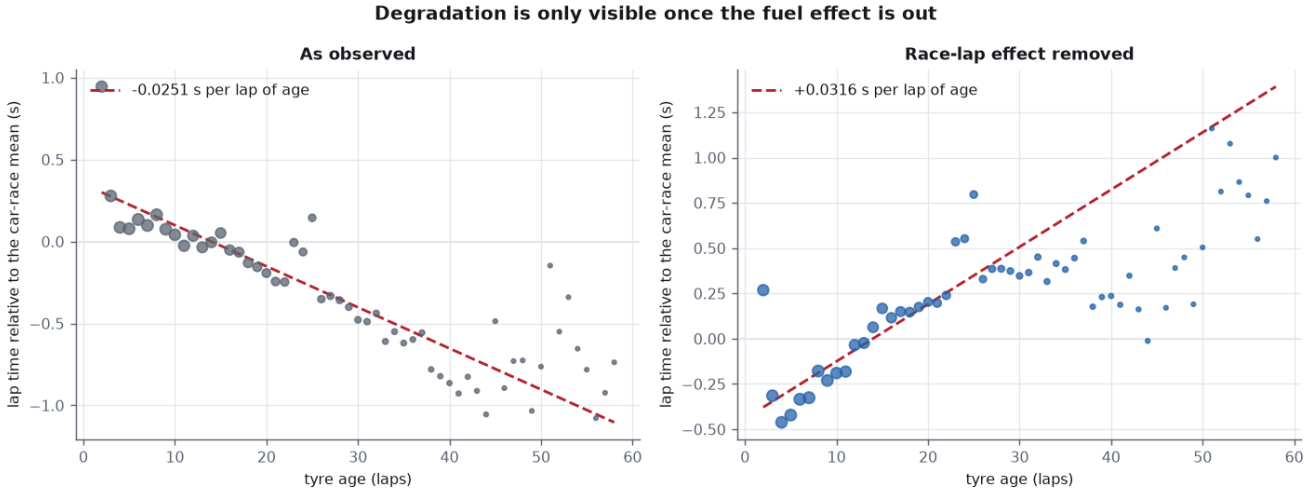


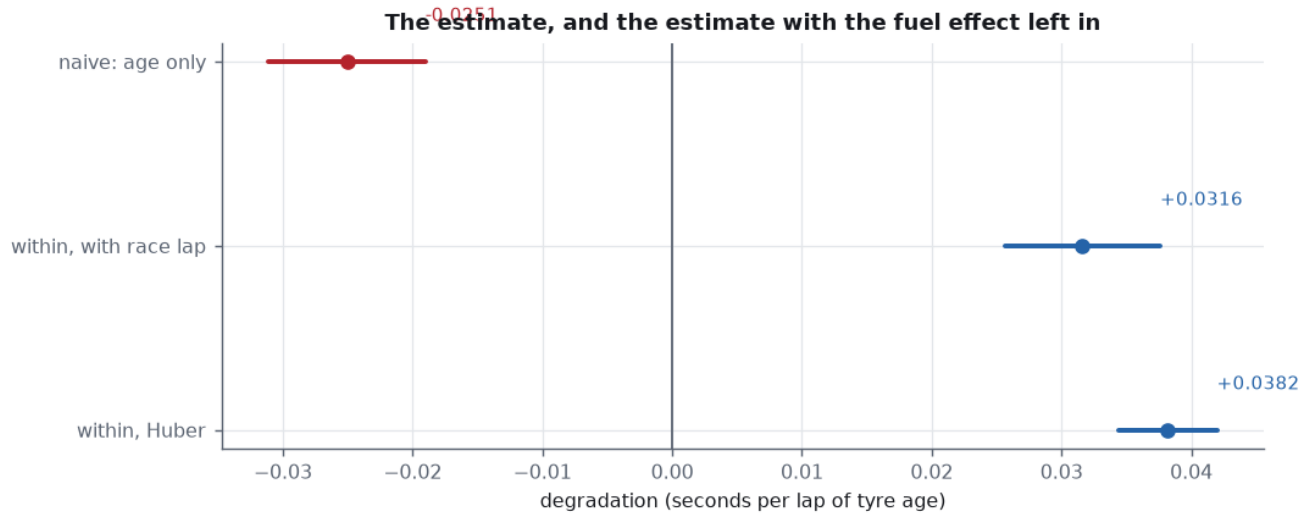
rule	laps	share
lap one	430	1.76%
neutralised	1,676	6.87%
in-lap	616	2.53%
out-lap	577	2.37%
retirement	116	0.48%
conditions	2,096	8.60%
short stint	106	0.43%
kept	18,769	76.97%

The estimate

Standard errors clustered on the car-race: 384 clusters, 18,769 laps. Within R² 0.1986, residual sigma 1.608 s.

Curvature in age is -0.00026 s per lap squared (p 0.124). The fixed effects are absorbed by the within transformation rather than by explicit dummies; the two agree to 3.6e-15 s, which is Frisch–Waugh–Lovell doing what it says.

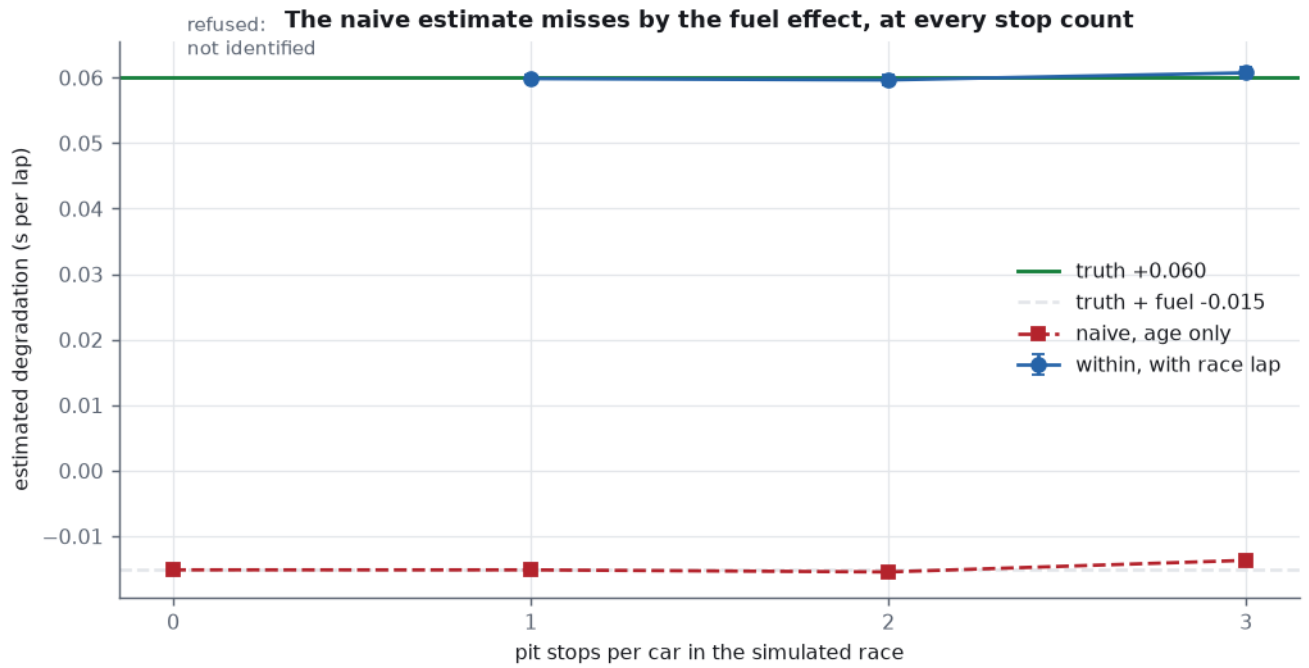




estimator	degradation, s/lap	std. error	95% interval	race-lap term
within, with race lap	+0.0316	0.0030	+0.0256 to +0.0376	-0.0546
within, Huber	+0.0382	0.0019	+0.0344 to +0.0420	-0.0532
naive, age only	-0.0251	0.0031	-0.0311 to -0.0190	—

Does the estimator recover a degradation it was given?

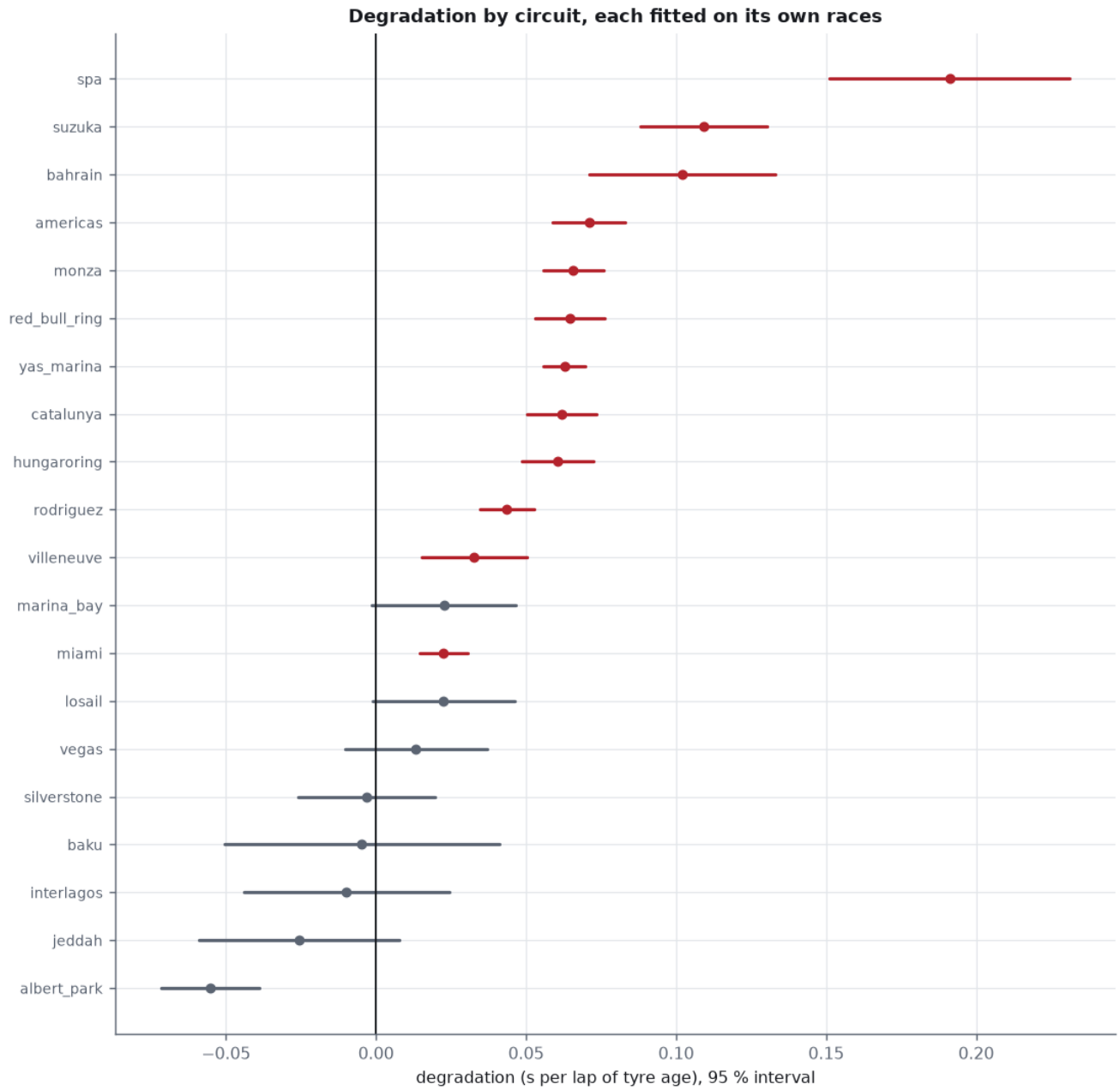
With no stops nothing identifies the split and the fitter refuses. The naive estimator never refuses, at any stop count, and is wrong by about the size of the fuel effect at all of them.



stops per car	true degradation	fitted degradation	fitted fuel	naive
0	+0.060	refused: not identified	—	-0.0151
1	+0.060	+0.0599	-0.0749	-0.0151
2	+0.060	+0.0597	-0.0751	-0.0155
3	+0.060	+0.0607	-0.0747	-0.0137

By circuit

12 of 20 circuits have an interval entirely above zero.

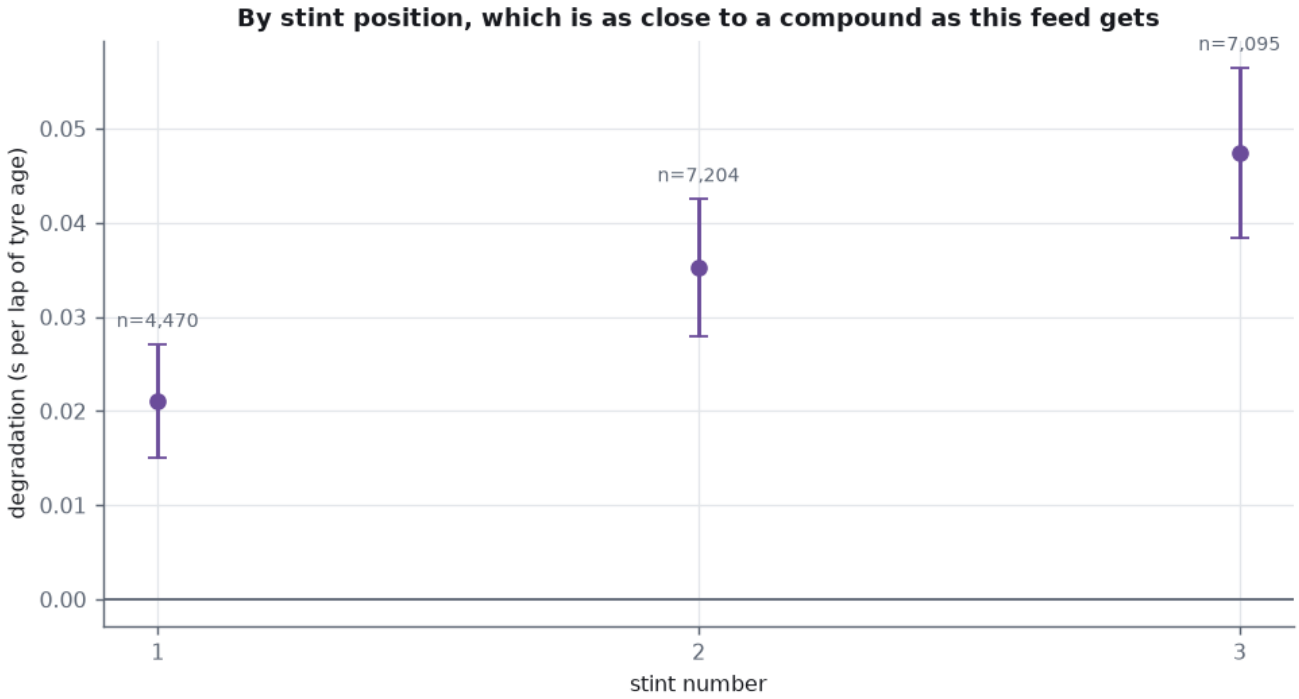


By circuit (continued)

circuit	degradation	95% interval	race-lap term	laps
spa	+0.1912	+0.1511 to +0.2312	-0.1070	711
suzuka	+0.1091	+0.0880 to +0.1303	-0.0735	702
bahrain	+0.1020	+0.0710 to +0.1329	-0.0687	915
americas	+0.0709	+0.0589 to +0.0830	-0.0654	913
monza	+0.0657	+0.0557 to +0.0757	-0.0340	883
red_bull_ring	+0.0646	+0.0530 to +0.0763	-0.0450	1,195
yas_marina	+0.0628	+0.0559 to +0.0697	-0.0603	1,058
catalunya	+0.0620	+0.0505 to +0.0736	-0.0580	1,196
hungaroring	+0.0606	+0.0488 to +0.0723	-0.0484	1,150
rodriguez	+0.0436	+0.0346 to +0.0526	-0.0355	1,124
villeneuve	+0.0328	+0.0154 to +0.0501	-0.0460	1,130
marina_bay	+0.0226	-0.0014 to +0.0466	-0.0289	977
miami	+0.0225	+0.0144 to +0.0307	-0.0512	1,071
losail	+0.0225	-0.0012 to +0.0463	-0.0966	830
vegas	+0.0133	-0.0102 to +0.0369	-0.0942	733
silverstone	-0.0032	-0.0260 to +0.0197	-0.0481	806
baku	-0.0047	-0.0506 to +0.0411	-0.0640	846
interlagos	-0.0099	-0.0441 to +0.0244	-0.0696	981
jeddah	-0.0256	-0.0591 to +0.0079	-0.0521	834
albert_park	-0.0553	-0.0716 to -0.0390	-0.0325	714

By stint position

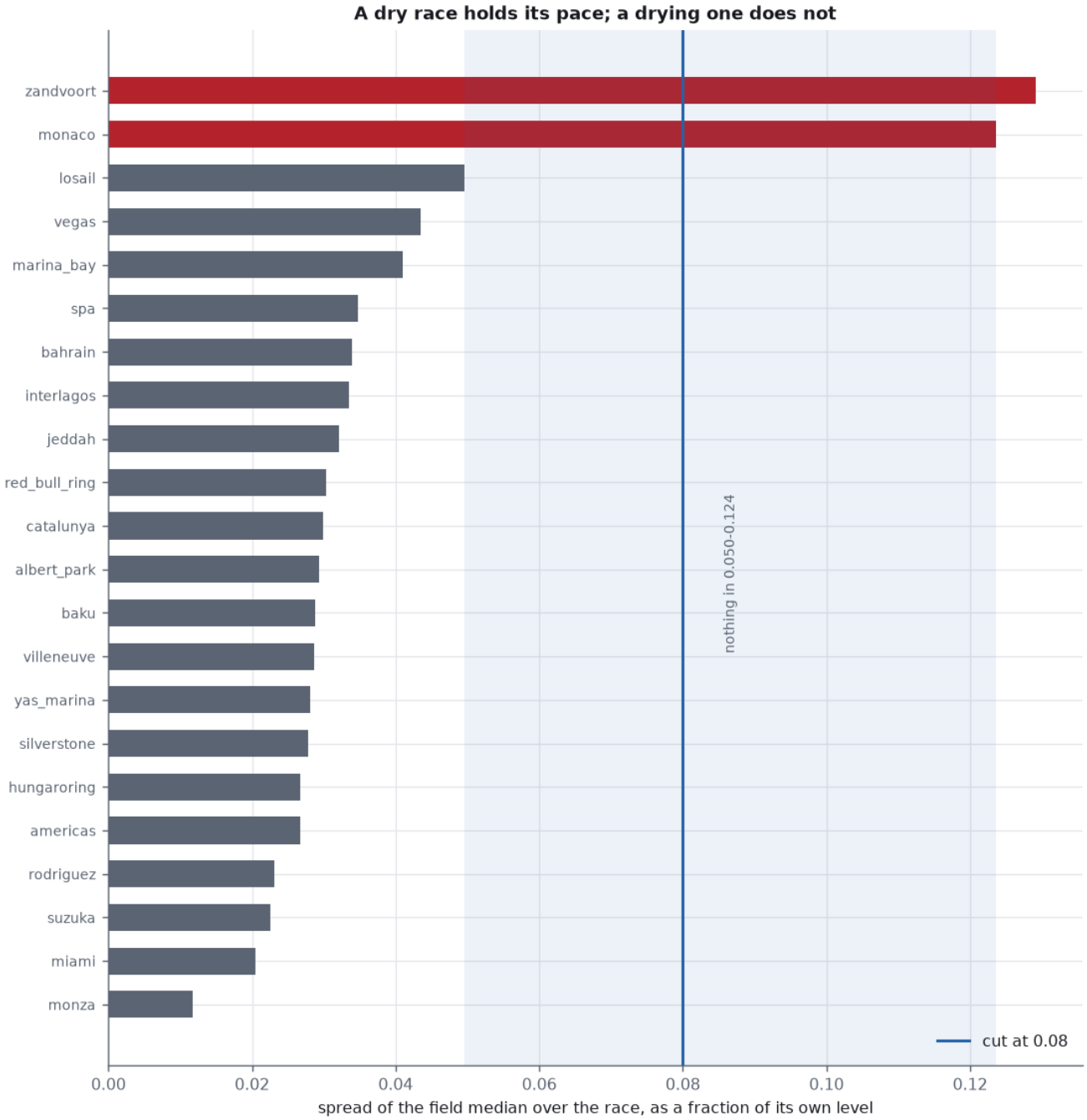
One model, with tyre age interacted with the stint number. This is as close to a compound breakdown as the feed allows: it never records what was fitted.



stint	degradation	95% interval	laps
stint 1	+0.0211	+0.0151 to +0.0271	4,470
stint 2	+0.0353	+0.0280 to +0.0426	7,204
stint 3	+0.0474	+0.0384 to +0.0565	7,095

The races that are not a race against the clock

A drying track makes the whole field faster lap after lap, which is a race-lap effect no fuel load can produce, and it swamps the model. The measure is the spread of the field median over the race. It runs from 0.012 to 0.050 for 20 races and then jumps to 0.124; the cut at 0.08 sits in that empty band, so anywhere in it removes the same races.

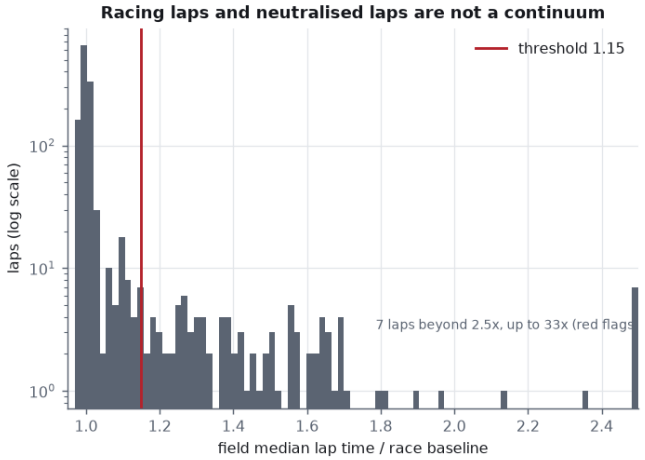


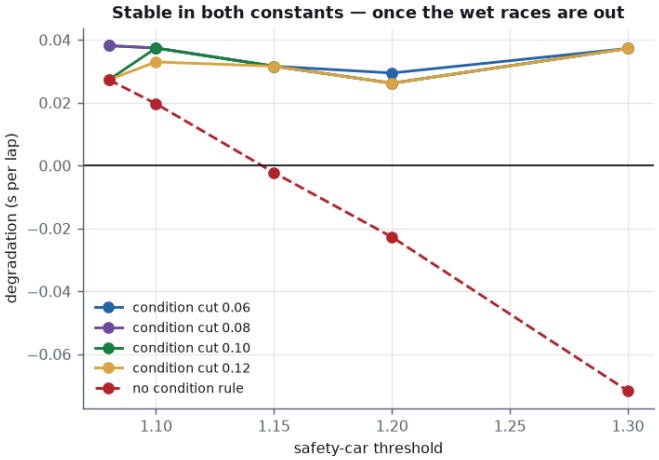
The races that are not a race against the clock (continued)

race	condition drift	verdict
zandvoort	0.129	excluded
monaco	0.124	excluded
losail	0.050	kept
vegas	0.044	kept
marina_bay	0.041	kept
spa	0.035	kept
bahrain	0.034	kept
interlagos	0.034	kept
jeddah	0.032	kept
red_bull_ring	0.030	kept
catalunya	0.030	kept
albert_park	0.029	kept
baku	0.029	kept
villeneuve	0.029	kept
yas_marina	0.028	kept
silverstone	0.028	kept
hungaroring	0.027	kept
americas	0.027	kept
rodriguez	0.023	kept
suzuka	0.023	kept
miami	0.021	kept
monza	0.012	kept

Do either of the cleaning constants matter?

With the condition rule on, the estimate moves by 0.0119 s per lap across every combination of the two constants. With it off, the safety-car threshold alone moves it by 0.0990 — more than the effect being measured. The fragile constant was never the safety-car rule; it was the two races the rule was being asked to paper over.

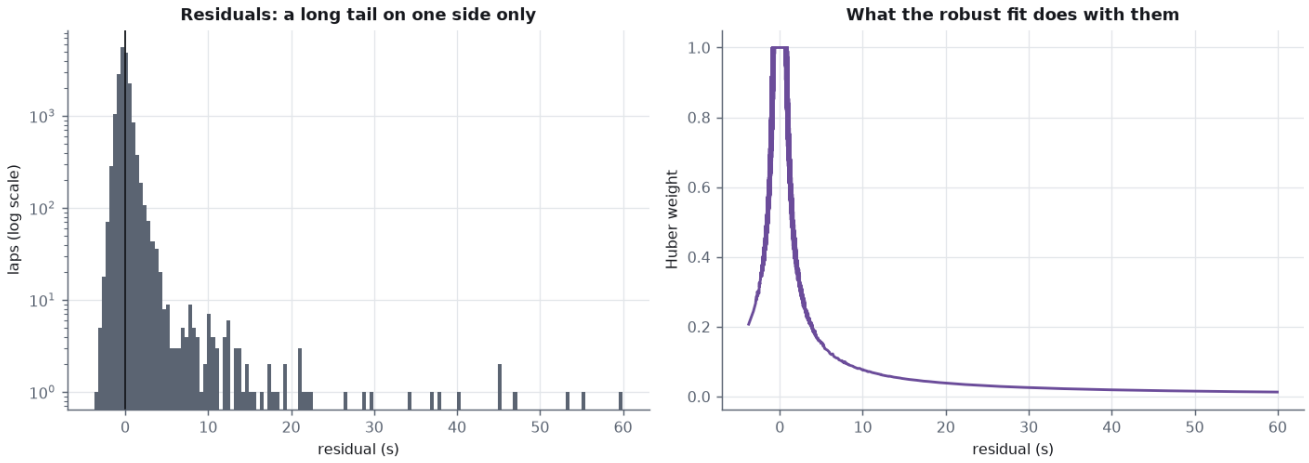




	cut 0.06	cut 0.08	cut 0.10	cut 0.12	no cut
safety car 1.08	+0.0382	+0.0382	+0.0273	+0.0273	+0.0273
safety car 1.10	+0.0374	+0.0374	+0.0374	+0.0330	+0.0197
safety car 1.15	+0.0316	+0.0316	+0.0316	+0.0316	-0.0022
safety car 1.20	+0.0295	+0.0262	+0.0262	+0.0262	-0.0227
safety car 1.30	+0.0373	+0.0373	+0.0373	+0.0373	-0.0717

Traffic, and why slow laps are not deleted

Being held up adds time to a lap and never subtracts it, so the residuals are skewed one way. The robust fit down-weights the far tail symmetrically; removing slow laps as a class would remove the end of every stint, which is where the degradation is.



Lap times, pit stops and results from the Jolpica mirror of the Ergast schema.